

SARTHAK DASH

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PROFESSIONAL SUMMARY

Photonics and optical researcher with 7+ years of experience across fiber laser development, visible laser engineering, and free-space optical communications (FSOC). Track record of quantified system-level outcomes, independent research, and publication in peer-reviewed journals (7) and conference proceedings (15).

TECHNICAL SKILLS

Free-Space Optical Comm. (FSOC): Optical terminal architecture, link budget analysis, SDA OCT / CCSDS standards, space-qualification evaluation

Pointing, Acquisition & Tracking: QPD signal chain design, FSM closed-loop control, acquisition algorithms, autocollimator-based alignment, satellite jitter analysis

Laser Systems & Amplifiers: Fiber laser design (CW, pulsed, Raman, supercontinuum), EDFA/MOPA chains, frequency conversion (SHG, BBO, LBO, PPLN), specialty fiber processing (stripping, cleaving, splicing, specialty fiber)

Laser Applications: Hyperspectral photoacoustic imaging, semiconductor wafer thermal monitoring, visible laser development, biomedical and industrial laser systems

Optical System Design: Multi-element optical train design and precision alignment, Zemax OpticStudio, COMSOL, atmospheric propagation modeling, free-space and nonlinear optics

Programming & Numerical Simulation: Python, Julia, MATLAB; ODE/PDE solvers, numerical optimization, Zemax scripting

Systems Engineering: Trade studies, first-principles analysis, vendor evaluation and qualification, requirements management, technical decision-making

Technical Leadership: Graduate student mentoring (4+), journal peer review (14+), graduate course instruction, cross-functional team collaboration

WORK EXPERIENCE

Qosmic satellite systems private limited | Founding Optical Engineer

Jan 2026 - Present

Free-space optical communications (FSOC) - satellite payload & ground station

- **Primary decision-maker for all optics-based design choices** across the FSOC system - transmitter chain, PAT subsystem, and ground-station receiver
- **Demonstrated terrestrial free-space optical link** over ~3 km through real atmosphere - primary architect of end-to-end PAT acquisition and communications validation testbed
- **Designed Pointing, Acquisition & Tracking (PAT) system** - acquisition architecture, QPD/FSM closed-loop tracking with sub- μ rad precision
- **Selected and designed laser assemblies** for both space and ground segments: communication transmitter (space qualified) and high-power uplink beacon; drove end-to-end vendor evaluation and space-qualification decisions
- **Designed internal simulation tools** for end-to-end link budget and SNR/BER trade studies, PAT algorithms and turbulence effects

Indian Institute of Science (IISc) | Sr. Research Fellow

Mar 2025 - Dec 2025

High-power visible laser development and graduate mentoring

- **Worked on extending cascaded Raman laser systems** into the blue-violet spectral window - a wavelength regime unavailable previously in this architecture - enabling new high-power visible source capabilities for spectroscopy and imaging applications

Indian Institute of Science (IISc) | Project Associate

Aug 2024 - Mar 2025

Frequency-converted tunable visible laser system

- **Achieved frequency-doubled, single-mode visible laser** wavelength-tunable from 530 - 720 nm (>100 mW in green, ~70 mW in yellow, ~55 mW in red and <100 GHz linewidth) for biomedical imaging and industrial diagnostics - best-in-class output properties with tuning

- **Developed novel cascaded Raman fiber lasers** achieving 5× improved spectral density and >99% spectral purity, establishing tunable high-power sources across the near-infrared window
- **Designed and optimized a supercontinuum source** for multi-point silicon wafer thermal monitoring system achieving $\pm 1^\circ\text{C}$ accuracy (best-in-class) leading to patent filing in collaboration with a leading semiconductor equipment manufacturer
- **Built tunable pulsed NIR/visible laser platform for hyperspectral photoacoustic imaging**, designed the nonlinear frequency conversion chains spanning NIR-to-visible
- **Published 6+ peer-reviewed journal articles** (Optics Letters, Optics Express) and presented at 12+ international conferences including SPIE Photonics West (Travel Grant awardee)
- **Mentored 4 graduate students** in laser development and system optimization; assisted in 2 graduate-level courses; reviewed 4+ manuscripts for top-tier photonics journals

EDUCATION

PhD | Indian Institute of Science (IISc), Bengaluru

Aug 2018 - Oct 2024

Thesis: Architectures for Linewidth Reduction in Cascaded Raman Fiber Lasers and Applications

Supervisor: Prof. V.R. Supradeepa

M.Sc. in Physics | NIT Rourkela

2015 - 2017

Focus: Nonlinear Dynamics and Numerical Simulation (Python, Julia, MATLAB)

ACHIEVEMENTS & AWARDS

- SPIE Photonics West "Fiber Lasers" Travel Grant (2024, 2025)
- SERB International Travel Support (ITS) (2024)
- Best Oral Presentation Award - 11th CeNSE Student Research Symposium (2024)
- National-level examinations: GATE (AIR 126, 2018), NET (2017), IIT-JAM (AIR 501, 2015)

PUBLICATIONS

PEER-REVIEWED JOURNAL ARTICLES

1. **S. Dash**, R. Deheri, V. Choudhury, and V. Supradeepa, "Fourier spectral shaper assisted feedback for wavelength and linewidth control of cascaded Raman fiber lasers," [Optics Letters 50\(1\), 201-204 \(2025\)](#).
2. **S. Dash**, R. Deheri, and V. Supradeepa, "Linewidth reduced cascaded Raman fiber lasers and their harmonic conversion for visible laser sources," [Optics Express 32\(12\), 20629–20637 \(2024\)](#).
3. R. Deheri, **S. Dash**, V. Supradeepa, and V. Balaswamy, "Cascaded Raman fiber lasers with ultrahigh spectral purity," [Optics Letters 47\(14\), 3499–3502 \(2022\)](#).
4. A. Goswami*, **S. Dash***, S. Avasthi, and V. Supradeepa, "Contactless temperature measurement of in-process silicon wafer using a spectrally-shaped supercontinuum source," [Optics Express, 33\(9\), 19677-19688 \(2025\)](#). (*equal contribution)
5. A. Goswami, S. Padmanavan, **S. Dash**, J. Prakash, and V. Supradeepa, "Pulsed cascaded Raman fiber laser widely tunable in the second near-infrared and visible window for hyperspectral photoacoustic imaging," [Optics Letters, 50\(7\), 2223-2226 \(2025\)](#).
6. S. Arora, S. Pal, C. Lakshmi, **S. Dash**, V. Supradeepa "Frequency comb-based seed laser architecture with improved Brillouin performance for spectral beam combining of narrow-linewidth lasers," [IEEE Photonics Journal, 17 \(3\), 1-6 \(2025\)](#).
7. S. Pal, S. Arora, **S. Dash**, C. Lakshmi, V. Supradeepa, "Brillouin-assisted generation and de-multiplexing of widely tunable high-repetition rate 1064 nm optical frequency combs with applications in spectral beam combining," [Optics Express 33\(10\), 21105-21115 \(2025\)](#).

CONFERENCE PROCEEDINGS

1. **S. Dash**, R. Deheri, and V. R. Supradeepa, "Continuously tunable green to red visible laser sources through harmonic conversion of cascaded Raman fiber lasers," Proc. SPIE 13342, Fiber Lasers XXII: Technology and Systems, 133420G (2025)

2. A. Goswami, S. Padmanavan, **S. Dash**, J. Prakash, and V. Supradeepa, "Passively Q-switched Raman fiber laser source widely tunable in near infrared and visible window," Proc. SPIE 13347, Nonlinear Frequency Generation and Conversion: Materials and Devices XXIV, 133470T (2025)
3. A. Goswami, S. Padmanavan, **S. Dash**, J. Prakash, and V. Supradeepa, "Widely tunable Raman fiber laser for hyperspectral photoacoustic imaging," Proc. SPIE 13319, Photons Plus Ultrasound: Imaging and Sensing 2025, 1331911 (2025).
4. **S. Dash**, R. Deheri, and V. R. Supradeepa, "High power, widely tunable, near-infrared and visible laser sources using Raman fiber lasers," in *2024 Conference on Lasers and Electro-Optics Pacific Rim (CLEO-PR)*, Technical Digest Series (Optica Publishing Group, 2024), paper Mo2H_4.
5. **S. Dash**, R. Deheri, and V. Supradeepa, "Linewidth control of cascaded Raman fiber lasers and visible conversion," Proc. SPIE 12865, Fiber Lasers XXI: Technology and Systems, 128650N (2024).
6. **S. Dash**, A. Goswami, R. Deheri, S. Avasthi, and V. Supradeepa, "Multi-point thermal monitoring of silicon wafer under processing utilizing a spectrally shaped supercontinuum source," Proc. SPIE 12878, High-Power Laser Materials Processing: Applications, Diagnostics, and Systems XIII, 128780I (2024).
7. A. Goswami, **S. Dash**, R. Deheri, and V. Supradeepa, "Widely tunable visible pulsed cascaded Raman fiber laser source," Proc. SPIE 12869, Nonlinear Frequency Generation and Conversion: Materials and Devices XXIII, 128690H (2024).
8. R. Deheri, **S. Dash**, V. Supradeepa, and V. Balaswamy, "Experimental Investigation on Linewidth Evolution of Cascaded Raman Fiber Laser Pumped with Low Intensity Noise Fiber Amplifiers," *Advances in Photonics Integrated Circuits, LASER and Applications. PHOTONICS* 2023.
9. A. Goswami, **S. Dash**, R. Deheri, V. Supradeepa, "Widely Tunable Visible Source Using Pulsed Cascaded Raman Fiber Laser," *Advances in Photonics Integrated Circuits, LASER and Applications. PHOTONICS* 2023.
10. S. Arora, **S. Dash**, S. Pal, C. Lakshmi, V. Supradeepa, "Spectral Beam Combining of Narrow-Linewidth Lasers from a Phase Modulated Frequency Comb-Based Seed Source," *Advances in Photonics Integrated Circuits, LASER and Applications. PHOTONICS* 2023.
11. R. Deheri, **S. Dash**, V. Supradeepa, and V. Balaswamy, "Cascaded Raman fiber lasers pumped with narrow linewidth, low intensity noise sources," in *Conference on Lasers and Electro-Optics/Pacific Rim* (Optica Publishing Group, 2022), p. CTuP1D_03.
12. A. Goswami, **S. Dash**, R. Deheri, S. Arun, and V. Supradeepa, "Pulsed cascaded Raman fiber laser with wide wavelength tunability," in *Conference on Lasers and Electro-Optics/Pacific Rim* (Optica Publishing Group, 2022), p. CTuP1D_02.
13. **S. Dash**, R. Deheri, V. Choudhury, and V. Supradeepa, "Fourier pulse shaper assisted feedback in cascaded Raman lasers for reduced linewidth and wide wavelength tunability Proc. SPIE 11997, Optical Components and Materials XIX, 119970K (2022).
14. R. Deheri, **S. Dash**, V. Supradeepa, and V. Balaswamy, "Cascaded Raman fiber lasers with very high spectral purity and low intensity noise," Proc. SPIE 11981, Fiber Lasers XIX: Technology and Systems, 119810K (2022).
15. **S. Dash**, R. Deheri, V. Choudhury, and V. Supradeepa, "Tunable random distributed feedback Raman fiber laser with Fourier spectral shaper for feedback control," Proc. SPIE 11665, Fiber Lasers XVIII: Technology and Systems, 116650Q (2021).